# Guided Problem Solving - 4

## Foundations of Fluid Mechanics, AM2530

September 15, 2026

1. A small lawn sprinkler is shown. The sprinkler operates at a gage pressure of 140 kPa. The total flow rate of water through the sprinkler is 4 L/min. Each jet discharges at 17 m/s (relative to the sprinkler arm) in a direction inclined 30° above the horizontal. The sprinkler rotates about a vertical axis. Friction in the bearing causes a torque of 0:18 N.m opposing rotation. Evaluate the torque required to hold the sprinkler stationary

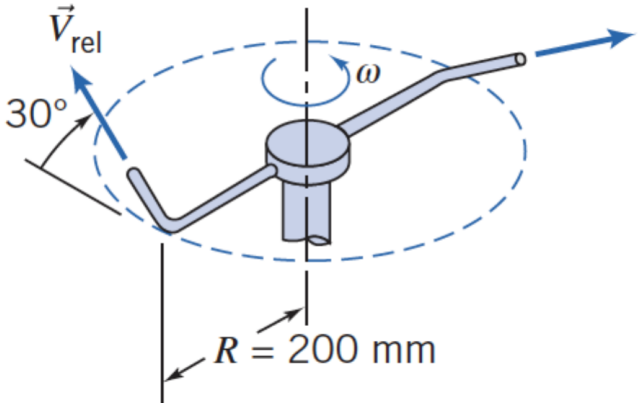



Figure 1: Lawn sprinkler

2. A jet strikes a vane that moves to the right at constant velocity Vc on a frictionless cart. Compute (a) the force Fx required to restrain the cart and (b) the power delivered to the cart. Also find the cart velocity for which (c) the force Fx is a maximum and (d) the power is a maximum.

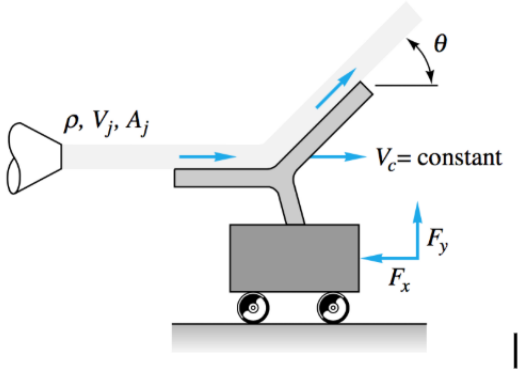



Figure 2

3. A ball is suspended in an air jet in a stable position as shown in the figure. Experiments show that the equilibrium height $h$ of the ball is found to depend on the ball diameter $D$ and weight $W$, jet diameter $d$ and velocity $V$, and the air density $\rho$ and viscosity $\mu$. Determine the dimensionless parameters that characterize this experiment.

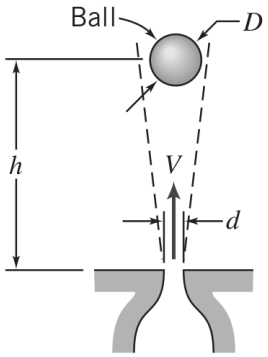



Figure 3

4. In some speed ranges, vortices are shed from the rear of bluff cylinders placed across a flow. The vortices alternately leave the top and bottom of the cylinder, as shown, causing an alternating force normal to the freestream velocity. The vortex shedding frequency, $f$, is thought to depend on $\rho$, $d$, $V$, and $\mu$. Use dimensional analysis to develop a functional relationship for $f$. Vortex shedding occurs in standard air on two cylinders with a diameter ratio of 2. Determine the velocity ratio for dynamic similarity, and the ratio of vortex shedding frequencies.

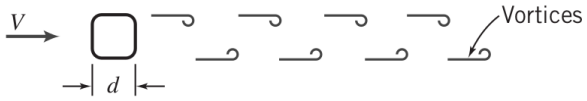



Figure 4

5. An airplane wing, with chord length of 1.5 m and span of 9 m is designed to move through standard air at a speed of 7.5 m/s. A 1 : 10 scale model of this wing is to be tested in a water tunnel. Determine the speed necessary in the water tunnel to achieve dynamic similarity. Determine the ratio of forces measured in the model flow to those on the prototype wing.

   *If these dimensionless coefficients are identical for both the model and the prototype, the solutions to the differential equations will be identical. The flows are then dynamically similar.*

6. When a small tube is dipped into a pool of liquid, surface tension causes a meniscus to form at the free surface, which is elevated or depressed depending on the contact angle at the liquid–solid–gas interface. Experiments indicate that the magnitude of this capillary effect, $\Delta h$, is a function of the tube diameter, $D$, liquid specific weight, $\gamma$, and surface tension, $\sigma$. Determine the number of independent $\Pi$ parameters that can be formed and obtain a set.

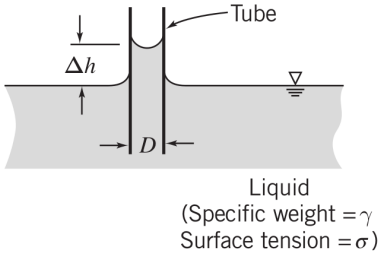



Figure 5